

Model Question Paper-I with effect from 2023-24 (CBCS Scheme)

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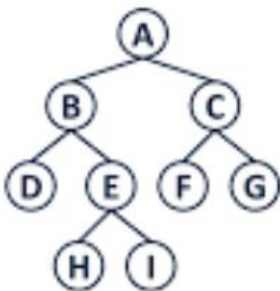
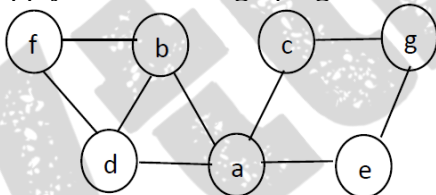
Third Semester B.E. Degree Examination Data Structures and Applications

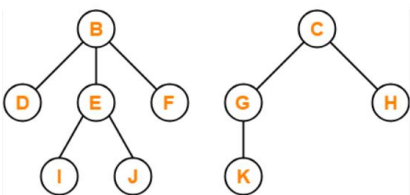
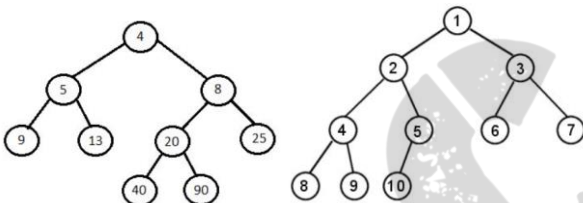
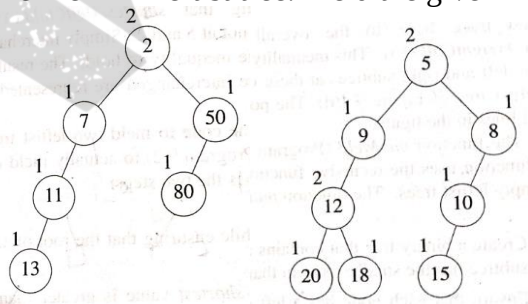
TIME: 03 Hours

Max. Marks: 100

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module -1			*Bloom's Taxonomy Level	Marks
Q.01	a	Define data structures. With a neat diagram, explain the classification of data structures with examples.	L2	5
	b	What do you mean by pattern matching? Outline the Knuth Morris Pratt (KMP) algorithm and illustrate it to find the occurrences of the following pattern. P: ABCDABD S: ABC ABCDAB ABCDABCDABDE	L3	8
	c	Write a program in C to implement push, pop and display operations for stacks using arrays.	L3	7
OR				
Q.02	a	Explain in brief the different functions of dynamic memory allocation.	L2	5
	b	Write functions in C for the following operations without using built-in functions i) Compare two strings. ii) Concatenate two strings. iii) Reverse a string	L3	8
	c	Write a function to evaluate the postfix expression. Illustrate the same for the given postfix expression: ABC-D*+E\$F+ and assume A=6, B=3, C=2, D=5, E=1 and F=7.	L3	7
Module-2				
Q. 03	a	Develop a C program to implement insertion, deletion and display operations on Linear queue.	L3	10
	b	Write a program in C to implement a stack of integers using a singly linked list.	L3	10
OR				
Q.04	a	Write a C program to implement insertion, deletion and display operations on a circular queue.	L3	10
	b	Write the C function to add two polynomials. Show the linked representation of the below two polynomials and their addition using a circular singly linked list P1: $5x^3 + 4x^2 + 7x + 3$ P2: $6x^2 + 5$ Output: add the above two polynomials and represent them using the linked list.	L3	10

Module-3																											
Q. 05	a	Write recursive C functions for inorder, preorder and postorder traversals of a binary tree. Also, find all the traversals for the given tree. 	L3	8																							
	b	Write C functions for the following i) Search an element in the singly linked list. ii) Concatenation of two singly linked list	L2	6																							
	c	Define Sparse matrix. For the given sparse matrix, give the linked list representation: $A = \begin{bmatrix} 0 & 0 & 3 & 0 & 4 \\ 0 & 0 & 5 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 6 & 0 & 0 \end{bmatrix}$	L3	6																							
OR																											
Q. 06	a	Write C Functions for the following i) Inserting a node at the beginning of a Doubly linked list Deleting a node at the end of the Doubly linked list	L3	8																							
	b	Define Binary tree. Explain the representation of a binary tree with a suitable example.	L2	6																							
	c	Define the Threaded binary tree. Construct Threaded binary for the following elements: A, B, C, D, E, F, G, H, I	L3	6																							
Module-4																											
Q. 07	a	Design an algorithm to traverse a graph using Depth First Search (DFS). Apply DFS for the graph given below. 	L3	8																							
	b	Construct a binary tree from the Post-order and In-order sequence given below In-order: GDHBAEICF Post-order: GHDBIEFCA	L2	6																							
	c	Define selection tree. Construct min winner tree for the runs of a game given below. Each run consists of values of players. Find the first 5 winners. <table><tr><td>10</td><td>9</td><td>20</td><td>6</td><td>8</td><td>9</td><td>90</td><td>17</td></tr><tr><td>15</td><td>20</td><td>20</td><td>15</td><td>15</td><td>11</td><td>95</td><td>18</td></tr><tr><td>16</td><td>38</td><td>30</td><td>25</td><td>50</td><td>16</td><td>99</td><td>20</td></tr></table>	10	9	20	6	8	9	90	17	15	20	20	15	15	11	95	18	16	38	30	25	50	16	99	20	L2
10	9	20	6	8	9	90	17																				
15	20	20	15	15	11	95	18																				
16	38	30	25	50	16	99	20																				

OR				
Q. 08	a	Define Binary Search tree. Construct a binary search tree (BST) for the following elements: 100, 85, 45, 55, 120, 20, 70, 90, 115, 65, 130, 145. Traverse using in-order, pre-order, and post-order traversal techniques. Write recursive C functions for the same.	L3	8
	b	Define Forest. Transform the given forest into a Binary tree and traverse using inorder, preorder and postorder traversal. 	L2	6
	c	Define the Disjoint set. Consider the tree created by the weighted union function on the sequence of unions: union(0,1), union(2,3), union(4,5), union(6,7), union(0,2), union(4,6), and union(0,4). Process the simple find and collapsing find on eight finds and compare which find is efficient.	L2	6
Module-5				
Q. 09	a	What is chained hashing? Discuss its pros and cons. Construct the hash table to insert the keys: 7, 24, 18, 52, 36, 54, 11, 23 in a chained hash table of 9 memory locations. Use $h(k) = k \bmod m$.	L3	10
	b	Define the leftist tree. Give its declaration in C. Check whether the given binary tree is a leftist tree or not. Explain your answer. 	L2	5
	c	What is dynamic hashing? Explain the following techniques with examples: i) Dynamic hashing using directories ii) Directory less dynamic hashing	L2	5
OR				
Q. 10	a	What is a Priority queue? Demonstrate functions in C to implement the Max Priority queue with an example. i) Insert into the Max priority queue ii) Delete into the Max priority queue iii) Display Max priority queue	L3	10
	b	Define min Leftist tree. Meld the given min leftist trees. 	L2	5
	c	Define hashing. Explain different hashing functions with examples. Discuss the properties of a good hash function.	L2	5